AutoCAD Report – Orthographic Projection of a Block

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1. Introduction

In this exercise, the block in Figure 3.51 was drawn in AutoCAD as three orthographic views: the elevation (front view), the plan (top view) and the left side view.

AutoCAD is a computer-aided design (CAD) program used by engineers and designers to make accurate 2D and 3D drawings. Instead of drawing by hand with a T-square and set squares, the user types commands and exact dimensions, so every line is drawn to its true size.

Orthographic projection shows a 3D object as several flat 2D views, each looking straight at one face of the object. The views are lined up with each other, so a point in one view can be projected to the other views with straight lines. In this drawing the plan is placed under the elevation and the left side view is placed to the right of the elevation, which is the first-angle projection layout.

The aim of the exercise is to read the dimensions from the isometric drawing, draw the three views to the correct size, show hidden edges with dashed lines, and add the main dimensions.

2. Description of the object

The object is a rectangular base block 75 mm long, 50 mm wide and 32 mm high, with a V-groove cut into its left end and a raised block on its right end. All sizes are in millimetres.

Feature	Dimensions (mm)
Base block	75 long × 50 wide × 32 high
V-groove (left end)	36 long; 9 flat on each side at the top; bottom of the V at 12 above the base, on the centre line (25 from the front)
Raised block (right end)	24 long × 24 wide × 24 high, set 12 back from the front face, with one sloping face running down to the top of the base

The direction of viewing for the elevation is shown by the arrow F in Figure 3.51.

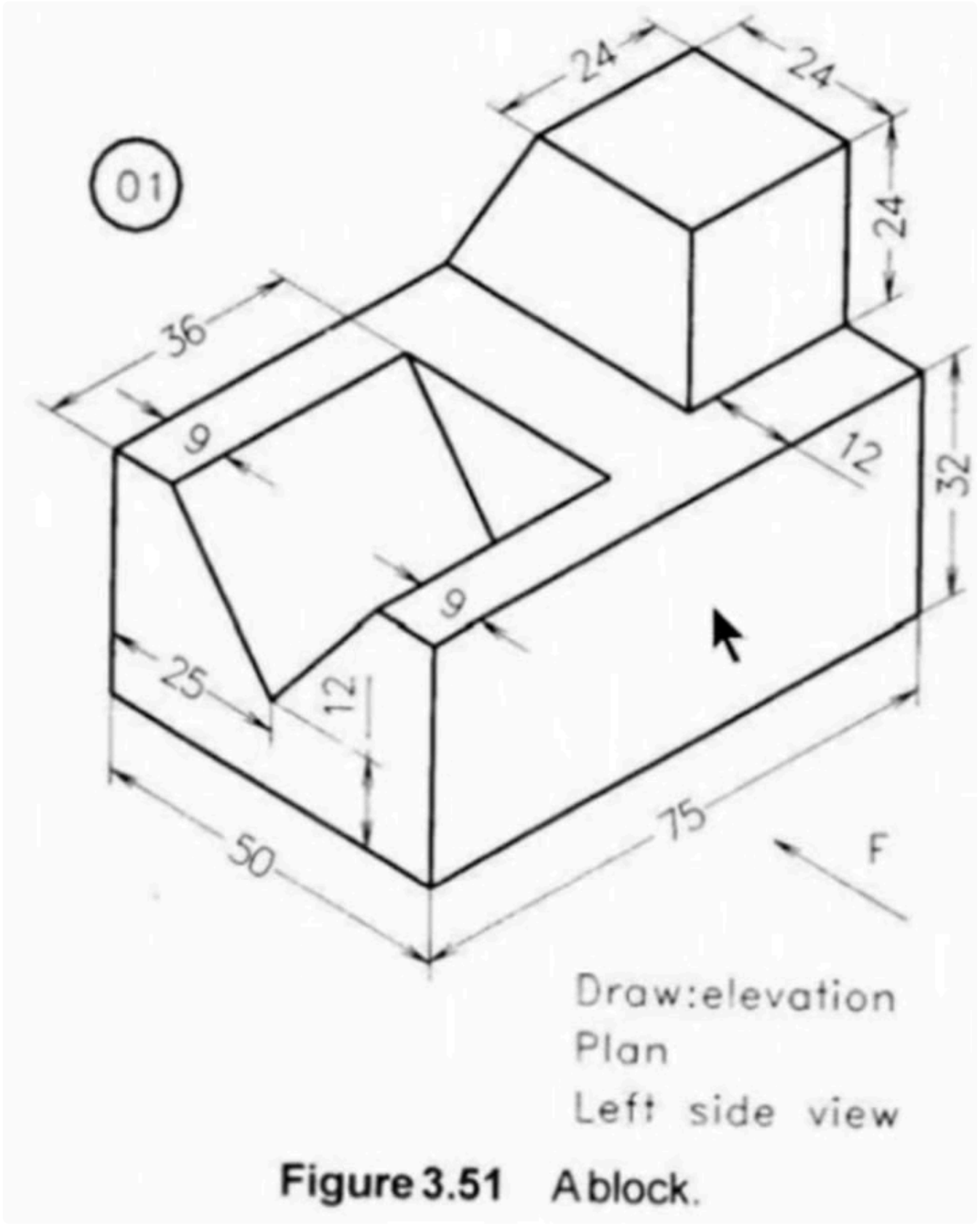


Figure 3.51 A block.

Figure 1 – The block in isometric view (Figure 3.51), with all its dimensions.

3. AutoCAD commands used

The drawing uses only straight lines, so it needs no CIRCLE, ARC or CENTER (centre line) commands: the object has no holes or curved surfaces.

Command (shortcut)	What it does in this drawing
UNITS (UN)	Sets the drawing units to millimetres and the number of decimal places
LIMITS	Sets the size of the drawing area
ORTHO (F8)	Forces lines to be exactly horizontal or vertical
OSNAP (F3)	Snaps the cursor to exact points such as endpoints, midpoints and intersections
LINE (L)	Draws the outlines of all three views and the projection lines
XLINE (XL)	Draws the long horizontal and vertical reference lines that divide the screen into four areas
OFFSET (O)	Copies a line at an exact distance, e.g. 9, 12, 24 or 36 mm
TRIM (TR)	Cuts away the parts of lines that are not needed
EXTEND (EX)	Lengthens a line until it meets another line
ERASE (E)	Deletes construction lines that are no longer needed
LAYER (LA)	Creates separate layers for visible lines, hidden lines and dimensions
LINETYPE / LTSCALE (LTS)	Loads the HIDDEN (dashed) line type and sets the size of its dashes
DIMLINEAR (DLI)	Adds horizontal and vertical dimensions
DIMSTYLE (D)	Sets the arrow size, text height, colour and decimal places of the dimensions
ZOOM (Z) / PAN (P)	Moves and enlarges the view on the screen

4. Drawing procedure

The views were drawn in this order: set-up, reference lines, elevation, plan, left side view, then hidden lines and dimensions.

Step 1 – Set up the drawing

1. Open a new drawing using the acadiso.dwt template (metric, millimetres).
2. Type UNITS and set the length type to Decimal and the units to millimetres.
3. Type LIMITS and set the lower-left corner to 0,0 and the upper-right corner to 297,210 (A4 size), then ZOOM → All.
4. Type LAYER and create three layers:
   - Visible – Continuous line type, black/white colour.
   - Hidden – HIDDEN line type (loaded with LINETYPE → Load).
   - Dimensions – Continuous line type, red colour.
5. Turn on ORTHO (F8) and OSNAP (F3) with Endpoint, Midpoint and Intersection selected.

Step 2 – Draw the reference lines

1. Use XLINE (or a long LINE) to draw one horizontal line and one vertical line that cross in the middle of the screen. They divide the screen into four areas.
2. The top-left area is for the elevation, the bottom-left area for the plan, and the top-right area for the left side view.
3. From the point where the two lines cross, draw a line at 45° into the bottom-right area: LINE, pick the intersection, then type @100<-45. This 45° mitre line is used to transfer the width (depth) of the object from the plan to the side view.

Step 3 – Draw the elevation (front view)

1. Set the Visible layer as current.
2. Draw the base as a rectangle 75 long and 32 high: LINE, pick a start point, then type @75,0 → @0,32 → @-75,0 → C (Close).
3. Draw the raised block on top of the right end: from the top-right corner of the base, draw a rectangle 24 long and 24 high (@0,24 → @-24,0 → @0,-24).
4. The V-groove cannot be seen from the front, so it is drawn as hidden lines: OFFSET the left edge by 36 and the bottom edge by 12, TRIM the extra parts, and move these two lines to the Hidden layer.

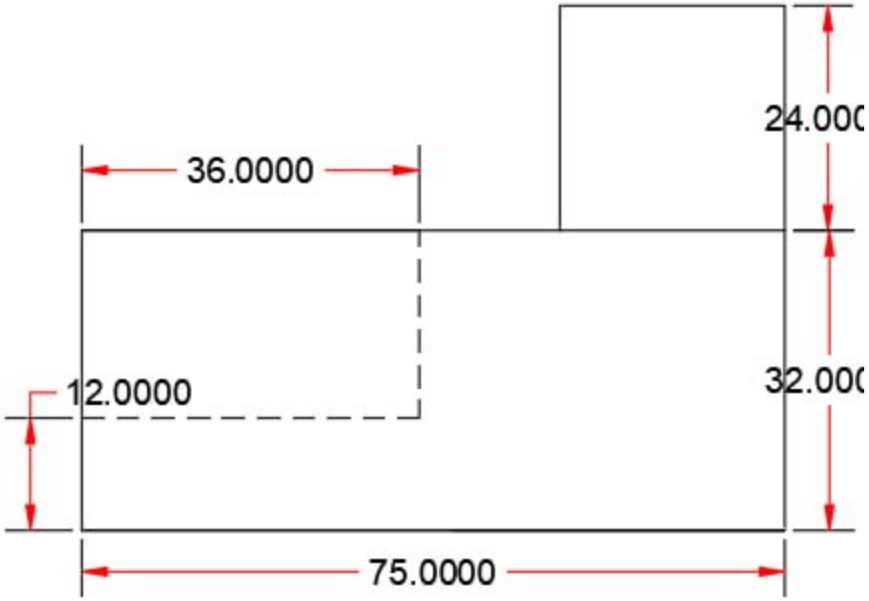


Figure 2 – The elevation (front view): the 75 × 32 base, the 24 × 24 raised block, and the V-groove shown as dashed hidden lines (36 long, 12 above the base).

Step 4 – Draw the plan (top view)

1. From the corners of the elevation, draw vertical projection lines down into the plan area.
2. Draw the outline of the plan as a rectangle 75 long and 50 wide, between the two outer projection lines.
3. Draw the V-groove: OFFSET the front and back edges by 9 to get the two top edges of the groove, and draw a line on the centre line (25 from the front) for the bottom of the V. Limit these lines to the 36 mm length of the groove with TRIM.
4. Draw the raised block: a rectangle 24 long at the right end, set back 12 from the front edge, with one more line where the sloping face meets the top face.
5. ERASE or TRIM the projection lines that are not part of the view.

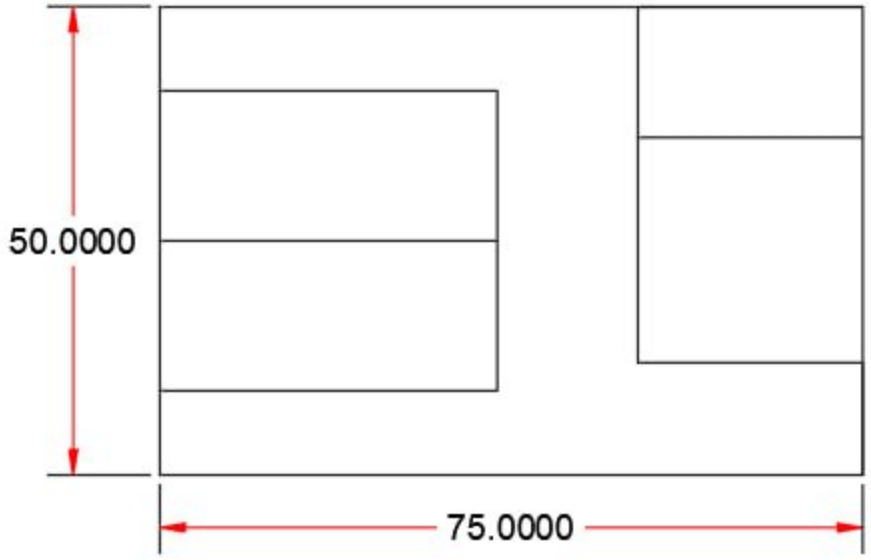


Figure 3 – The plan (top view), looking down on the block: the 75 × 50 outline, the V-groove on the left with its centre line, and the raised block on the right.

Step 5 – Draw the left side view

1. From the top and bottom of the elevation, draw horizontal projection lines to the right into the side-view area. These give the heights (32 and 56).
2. From each point in the plan, draw horizontal lines to the right until they meet the 45° mitre line.
3. From the points where they meet the mitre line, draw vertical lines up into the side-view area. These give the widths (9, 25, 12, 24 and 50).
4. Join the points where the horizontal and vertical lines cross to draw the side view: the 50 × 32 base, the V-groove shape with its bottom 12 above the base, and the raised block with its sloping face.
5. TRIM and ERASE all construction lines.

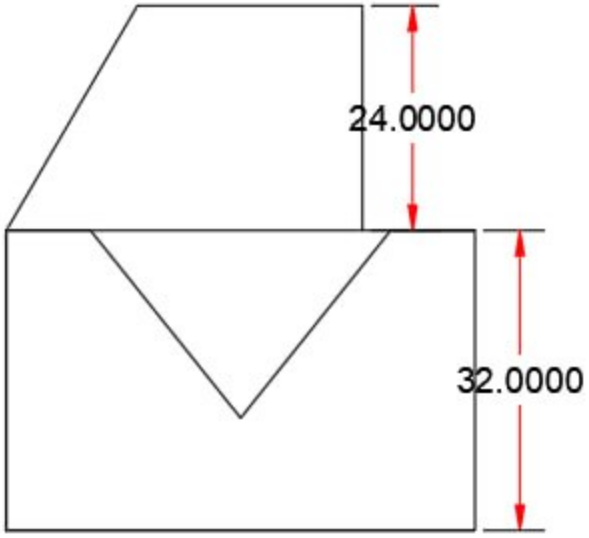


Figure 4 – The left side view: the 50 × 32 base with the V shape of the groove, and the raised block with its sloping face.

Step 6 – Check the hidden lines

1. Make sure every edge that is hidden behind the object is on the Hidden layer, so it appears dashed.
2. If the dashes are too small or too large, change them with LTSCALE (for example LTSCALE = 0.5).

Step 7 – Add the dimensions

1. Set the Dimensions layer as current.
2. Use DIMSTYLE to set the arrow size and text height (for example 2.5), and set the precision to 0 so dimensions show as 75 instead of 75.0000.
3. Use DIMLINEAR to add the main dimensions: 75 and 32 on the elevation, 24 for the raised block, 36 and 12 for the groove, 50 and 75 on the plan.
4. Place each dimension outside the view, clear of the lines of the object.

Step 8 – Save the drawing

1. Check that the three views line up with each other and that all sizes match Figure 3.51.
2. Save the file with SAVE AS (for example Block_3-51.dwg).

5. Conclusion

The block was drawn successfully in three orthographic views, with hidden lines and dimensions, using only basic AutoCAD commands.

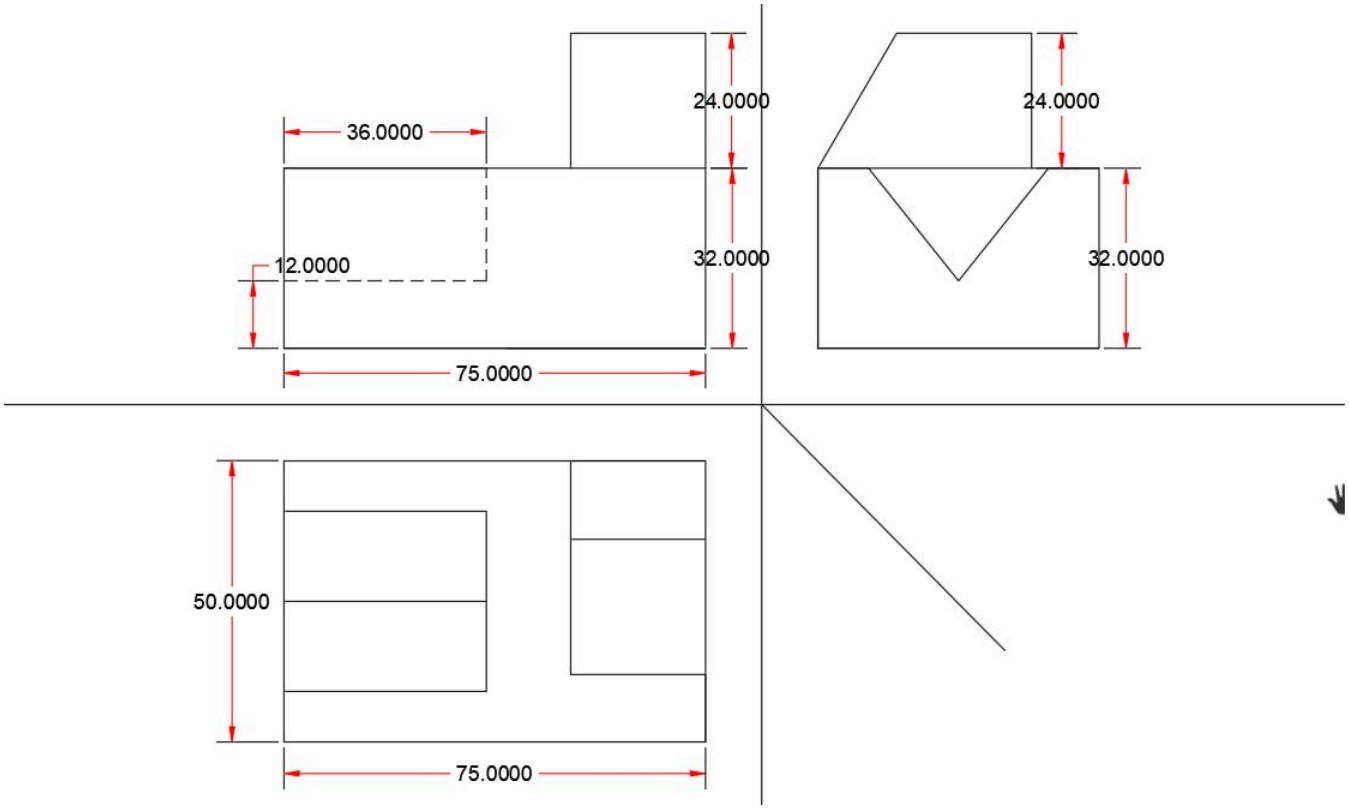


Figure 5 – The finished drawing: elevation (top left), left side view (top right), plan (bottom left) and the 45° mitre line (bottom right).

The exercise showed how projection lines and a 45° mitre line keep the elevation, plan and side view lined up, so each size only has to be measured once. Commands such as OFFSET and TRIM made the drawing faster and more accurate than drawing by hand, and layers made it easy to tell visible lines, hidden lines and dimensions apart.